DreamTask

ER: Requirements Specification Component

The purpose of this project is to develop a web application called DreamTask, which is a project management system that will be developed by a small software company, mainly to define and organize all possible tasks or goals to be achieved individually or as a team.

A1: Project Management

This App will be developed by a small team of software engineers, and will allow any type of user, whether corporate related like startups, consultants, small companies, large businesses, and many others, to have control of all their workflow. Any team of employees would benefit the most from this application since it will allow them to achieve professional success by controlling and planning their activities and by helping to build the right line-up in order to assemble the perfect project. In terms of operation, before the creation of a project, there has to be a workspace associated with that company, with all types of registered users: board members, administrators and people that might be relevant for the development of the project (observers and guests for instance). Each project has at least a team and task lists that are assigned to those members, tracking the one that defined that task as well as the one who completed it. Tasks inside the lists can be directed to groups and have upcoming deadlines and priority levels. Each team has a team coordinator. A search engine dedicated to finding the respective teams/tasks associated with each individual.

There is also a discussion forum to react to changes, proposals or issues. All types of information relevant to the individual or team, like a job promotion, an accomplished task, a new challenge in the workspace, or simply a new message in the forum, will be showcased as a notification, to reinforce the user's attention. Concerning the members of the workspace, authenticated users can have different permissions according to their privileges, whether being admins (the ones who have full responsibility over the management of the system) or regular members.

It is possible to change who adds members to a board: they can be selected by administrators only (with complete access and modification privileges that include adding and removing members) or by admins and members of the board. In both cases admins can add , edit and remove tasks by team members. Additionally, team members can switch projects in their workspace.

The commenting permission (deleting, adding and editing comments), on which the discussion forum mentioned above relies, is always controlled by admins. The default setting allows all members that are logged in to comment.

In terms of the design and structure of the application, the aim is to maintain an easy-to-use and transparent navigation system, accompanied by an appealing visual appearance to new visitors or potential customers of our product.

A2: Actors and User stories

This artifact contains the specification of the actors and their user stories, serving as agile documentation of the project’s requirements.

1. Actors

An Actor specifies a role played by a user or any other system that interacts with the subject. For the DreamTask the actors are represented in Figure 1 and described in Table 1.

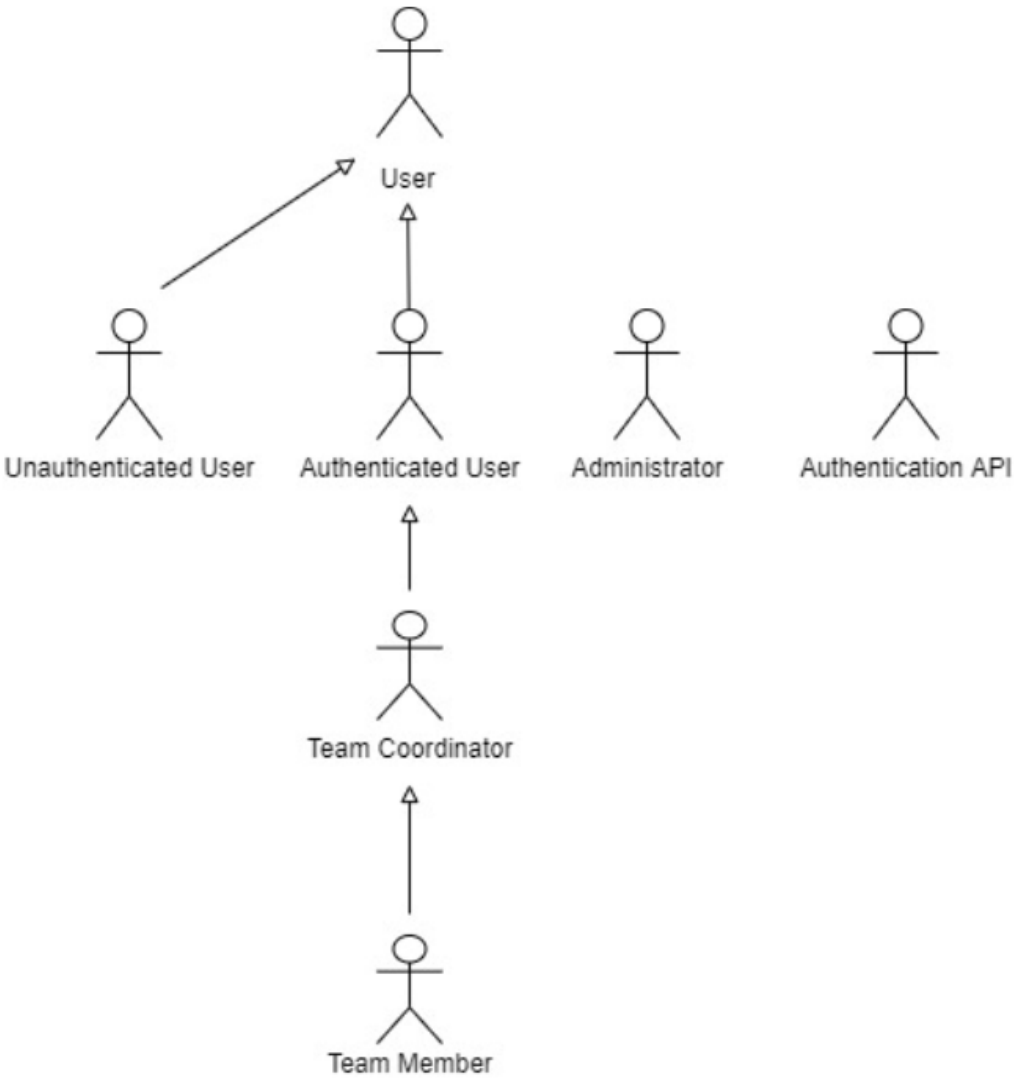


Figure 1: Actors

Identifier	Description	Examples
User	Generic User. Access to public information and can perform action as search and view pages.	n/a
Unauthenticated User	Can register or authenticate in the system.	n/a
Authenticated User	Can manage their personal information and do some actions in the system.	john123
Administrator	Responsible for the management, supervisory and moderation functions. Such as blocking users.	admin

Identifier	Description	Examples
Team member	Authenticated user; Project member allowed to complete, assign tasks and comment in team forum.	john123
Team Coordinator	Authenticated user responsible for projects; Allowed to add members to a project, create and distribute tasks to team members and moderate the team forum.	Boss1
Authentication API	External APIs that can be used to register and authenticate users into the system.	Google Sign In for Websites

Table 1: DreamTask actors descriptions

2. User Stories

A user story is a short, simple description of a feature told from the perspective of the person who desires the new capability, usually a user or customer of the system.

On the tables below are the user stories that define the functions of our application.

User

Identifier	Name	Priority	Description
US01	Homepage	high	As a User, I want to access the home page, so that I can see the website display and main layout.
US02	About page	high	As a User, I want to access the about page, to read a complete description of the website, learn about the history of the creators and find curiosities about the platform.
US03	FAQ page	high	As a User, I want to access the FAQ page so I can find solutions to my doubts in the form of objective questions to other users.
US04	Contacts page	high	As a User, I want to be able to contact the platform creators for any kind of pertinent question or seek help.

Table 2: Users user stories

Unauthenticated User

Identifier	Name	Priority	Description
US05	Sign-In	high	As an Unauthenticated User, I want to authenticate into the system, so that I can access my information.
US06	Sign-Up	high	As an Unauthenticated User, I want to register into the system, so that I can authenticate myself into the system.

Identifier	Name	Priority	Description
US07	Sign-In using external API	low	As an Unauthenticated User, I want to authenticate into the system using an external account, so that I can access my information.
US08	Sign-Up using external API	low	As an Unauthenticated User, I want to register into the system using an external account, so that I can authenticate myself into the system.

Table 3: Unauthenticated users user stories

Authenticated User

Identifier	Name	Priority	Description
US09	View profile	high	As an Authenticated User, I want to see my profile information, so that I can keep track of my info.
US10	Edit profile	medium	As an Authenticated User, I want to edit my profile, so that I can update my data.
US11	Delete account	high	As an Authenticated User, I want to delete my account, so that I can erase all my info from the website.
US12	Create Project	high	As an Authenticated User, I want to Create a Project, so that I can start my own project.
US13	Search Team	high	As an Authenticated User, I want to search for a team, so that I can find the team faster.
US14	Sign Out	high	As an Authenticated User, I want to sign out, so that I can leave the app safely.

Table 4: Authenticated users user stories

Administrator

Identifier	Name	Priority	Description
US15	Delete user	high	As an Admin, I want to be able to delete a user so that I can keep control of who is using the platform.
US16	Delete Team	high	As an Admin, I want to be able to delete a team so that I can keep control of what teams are using the platform.
US17	Add FAQ	low	As an Admin, I want to add the most common questions so the users don’t need to ask me the same things.
US18	Edit FAQ	low	As an Admin, I want to edit comments, so the I can manage the users comments and my own too.
US19	Remove FAQ	low	As an Admin, I want to remove the most common questions so the users don’t need to ask me the same things.

Table 5: Administrator's user stories

Team Member

Identifier	Name	Priority	Description
US20	Switch Project	high	As a Team Member, I want to change projects, so that I can always be satisfied and motivated with my work and still be inside the same workspace.
US21	Switch Team	high	As a Team Member, I want to switch teams, so that I can help a team that is having difficulties finishing their part of the project.
US22	View Projects	high	As a Team Member, I want to access the home page, so that I can see my projects.
US23	View Task	high	As an Authenticated User, I want to see my task, so that I can see more information about it like the description or the assignees.
US24	View TaskList	high	As a Team Member, I want to see my task list, so that I can see the amount of work I have to accomplish.
US25	Add Task	high	As a Team Member, I want to add a task, so that I can have the new task registered.
US26	Remove Task	high	As a Team Member, I want to remove a task so that whenever I can’t find its purpose on the project, it doesn’t affect the workflow.
US27	Search Task	high	As a Team Member, I want to search for a task, so that I can find the task faster.
US28	Add comment	high	As a Team Member, I want to comment in the team forum, so that I can inform others about proposals, errors or issues.
US29	Edit comment	high	As a Team Member, I want to edit a comment in the team forum, so that I can make clearer what I want to say.
US30	Delete my comment	high	As a Team Member, I want to eliminate a comment, so that I can have only constructive comments on the team forum.
US31	View history of previous tasks	high	As a Team Member, I want to view the history of previous tasks, so that I can recover lost tasks.
US32	Mark as Completed	low	As a Team Member, I want to mark the tasks as completed, so that I can keep track of the work done.
US33	Mark as Uncompleted	low	As a Team Member, I want to mark the tasks as completed, so that I can keep track of the unfinished work.
US34	Reply Comments	low	As a Team Member, I want to reply to comments, so that I can interact with a comment.

Table 6: Team Member's user stories

Team Coordinator

Identifier	Name	Priority	Description
US35	Distribute Tasks to Team Members	high	As a Team Coordinator, I want to distribute tasks to Team Members, so that I distribute tasks equally.
US36	Add Team Members	high	As a Team Coordinator, I want to add Team Members, so that I can have a bigger team.
US37	Remove Team Members	high	As a Team Coordinator, I want to remove Team Members, so that I can keep my Team Members updated.
US38	Edit Project Information	high	As a Team Coordinator, I want to Edit Project Information, so that I can change information about the project that needed to be improved.
US39	Remove Project	high	As a Team Coordinator, I want to Remove a Project, so that I can delete a project that is no longer needed.

Table 7: Team Coordinator's user stories

3. Supplementary requirements

The following section will present the chosen business rules, technical requirements and non-functional requirements on the project.

Business rules

A business rule defines the aspects and constraints of the business, with the purpose of implementing a certain business structure and influencing business actions.

Identifier	Name	Description
BS01	Task deadline	The tasks need to be completed within their respective deadline.
BS02	Task limit	The team member cannot add more than ten tasks in a work day.
BS03	Deleted Team Member Information	The deleted team member information has his data stored on the database but it is not visible to other users.

Table 8: Supplementary requirement´s Business Rules

Technical requirements

This part concerns technical aspects that will be used in the respective system, such as performance-related issues, reliability issues and availability issues.

Identifier	Name	Description

Identifier	Name	Description
TR01	App availability	The system must be available 99 percent of the time.
TR02	App accessibility	The system must provide the access of the pages to everyone ,independently of any condition, like being handicap, or the web browser they use.
TR03	Robustness	The app needs to continue execution, even when runtime errors occur.
TR04	Scalability	The app system has to be prepared to handle a big number of new users and their traffic.
TR05	Software ethics	Hidden user details or personal data, should not be shared without full acknowledgement and authorization from its owner.
TR06	Software security	The app needs to be prepared to protect information from external inputs or attacks that will try to corrupt the system data.
TR07	Data management	The database PostgreSQL database management system must be used, with a version of 11 or higher.
TR08	App portability	This system should be platform independent, making it available to the maximum number of users possible.

Table 9: Supplementary requirement´s Technical requirements

Restrictions

A restriction on the design puts barriers for the user

Identifier	Name	Description
C01	Project Deadline	The system should be ready to be used at the end of this semester.

Table 10: Supplementary requirement´s Restrictions

A3: Information Architecture

This artifact compacts the system information we are developing. The main goals are:

1.Identifying and describing the user requirements, and raise new ones

2.Preview and empirically test the user interface of the product to be developed.

3.Enable quick and multiple iterations on the design of the user interface.

This artifact includes two elements:

1.A sitemap, defining how the information is organized in pages.

2.A set of wireframes, defining the functionality and the content for each page. Wireframes are designed for at least two of the most important pages.

1. Sitemap

A sitemap is a visual representation of the relationship between the different pages of a website that shows how all the information fits together.

The DreamTask system is organized in seven main areas, the static pages that provide general information about the system (Static Pages), the individual unauthenticated user page (Unauthenticated User Pages), the pages with administration features (Admin Pages), the workspace page (Project Pages), the authenticated user pages (Authenticated User), the responsible for the team members' project pages (Team Coordinator Pages) and the team members' pages (Team Member Pages).

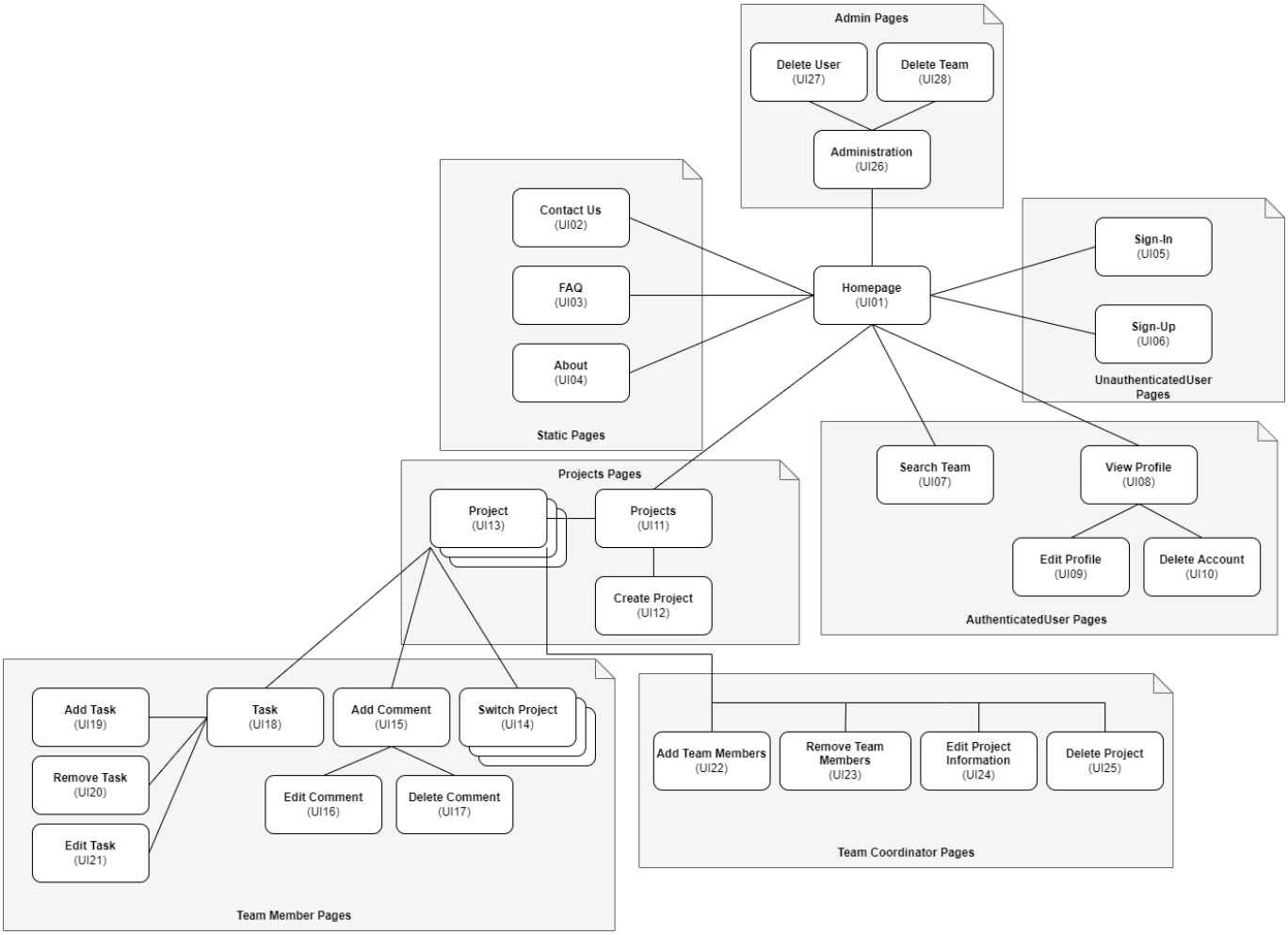


Figure 1: DreamTask Sitemap

2. Wireframes

Wireframe is a prototype used in interface design to suggest the structure of a website and the connection between its pages. A wireframe web is a similar illustration of the layout of fundamental elements in the interface.

2.1 Authenticated User Page

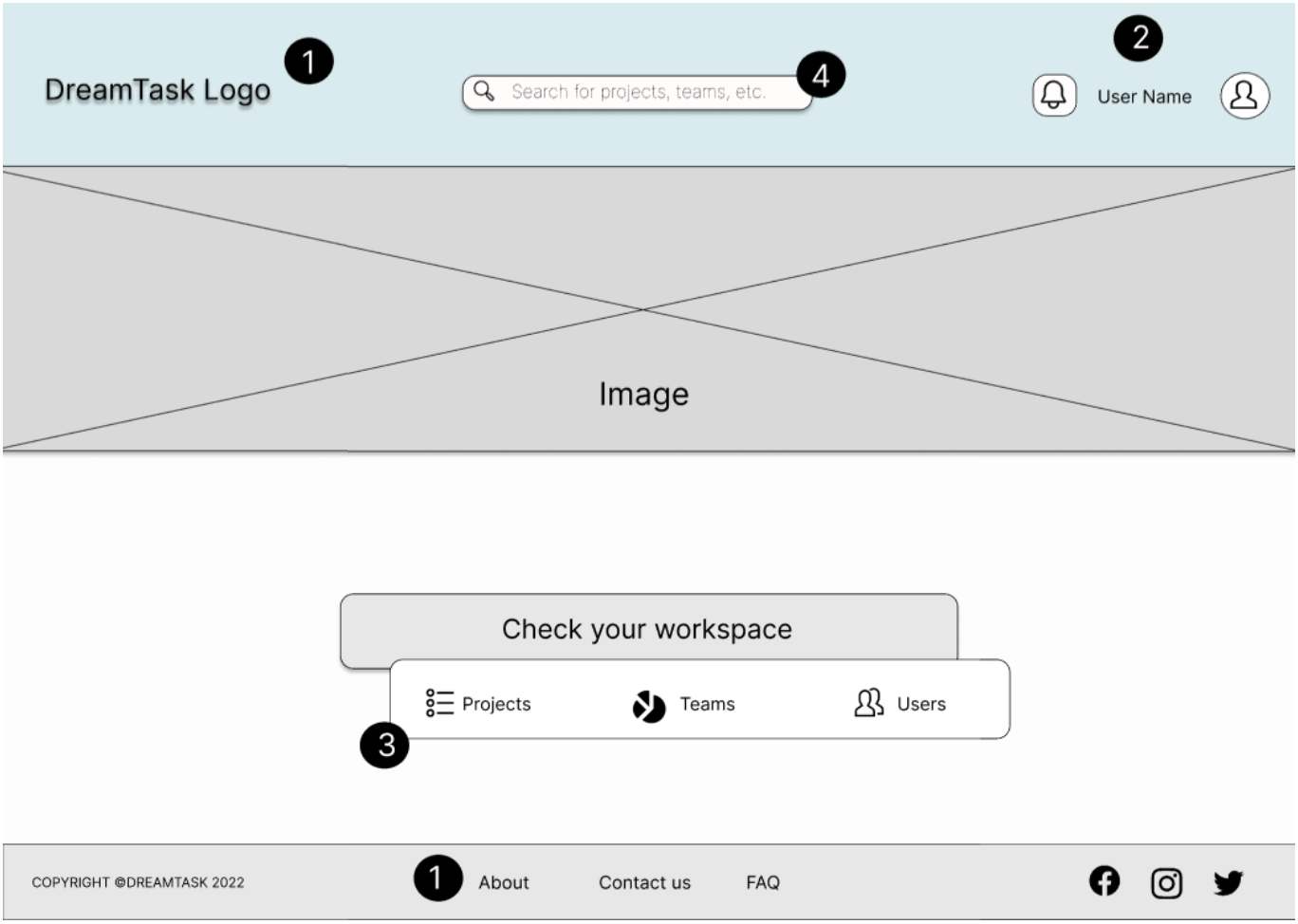


Figure 2: Authenticated User Wireframe

1. Navigation - logo as home button and another pages (about, contacts, FAQ)
2. User area (profile and notifications)
3. Navigation - redirect to workspace area with the projects, teams and users
4. Search Bar for projects, teams, users

2.2 Unauthenticated User Page

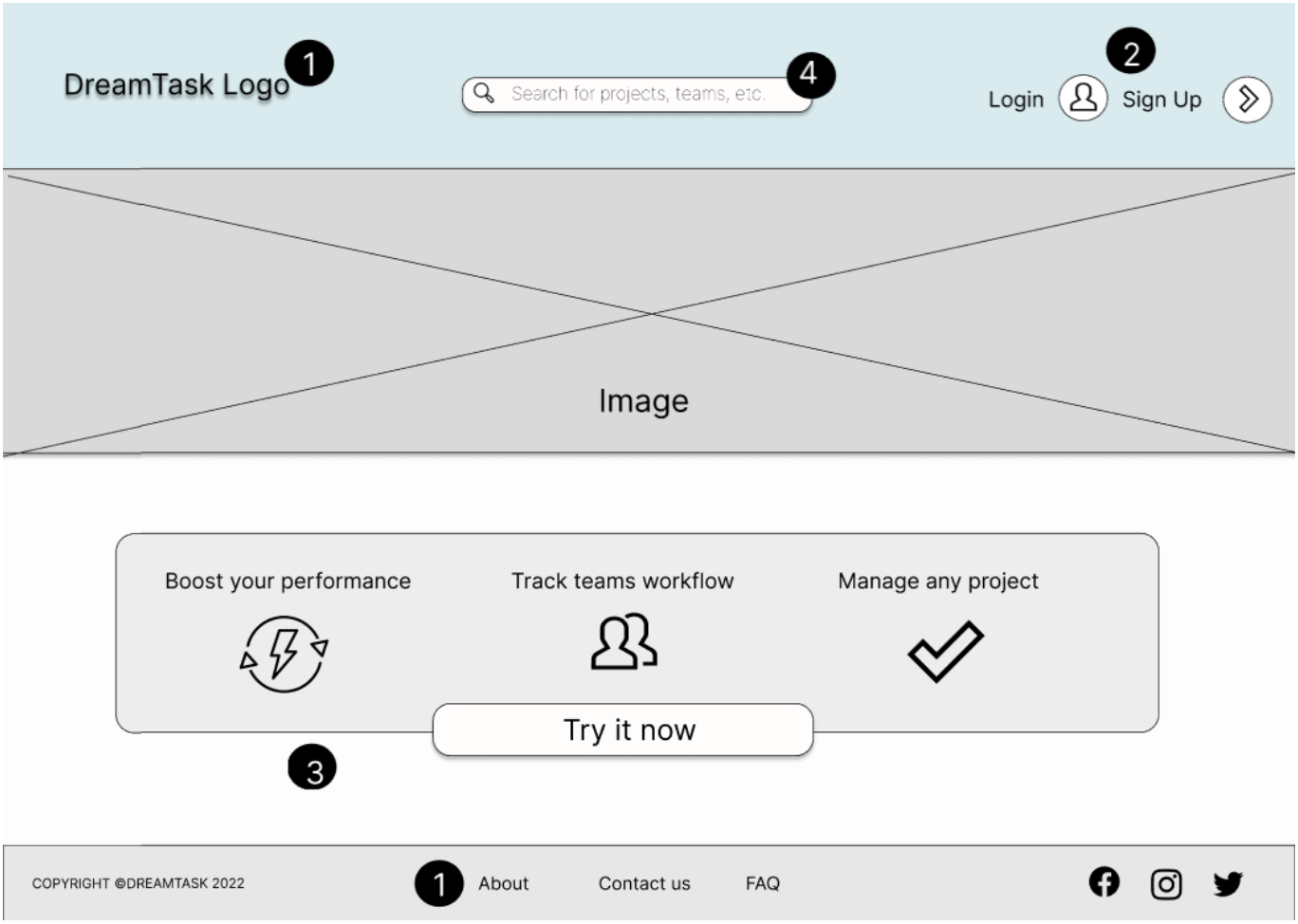


Figure 3: Unauthenticated User Wireframe

1. Navigation - logo as home button and another pages (about, contacts, FAQ)
2. Login or sign up pages redirection area
3. Redirection to About page with user app benefits and reasons to sign up
4. Search Bar for projects, teams, users

2.3 Login Page

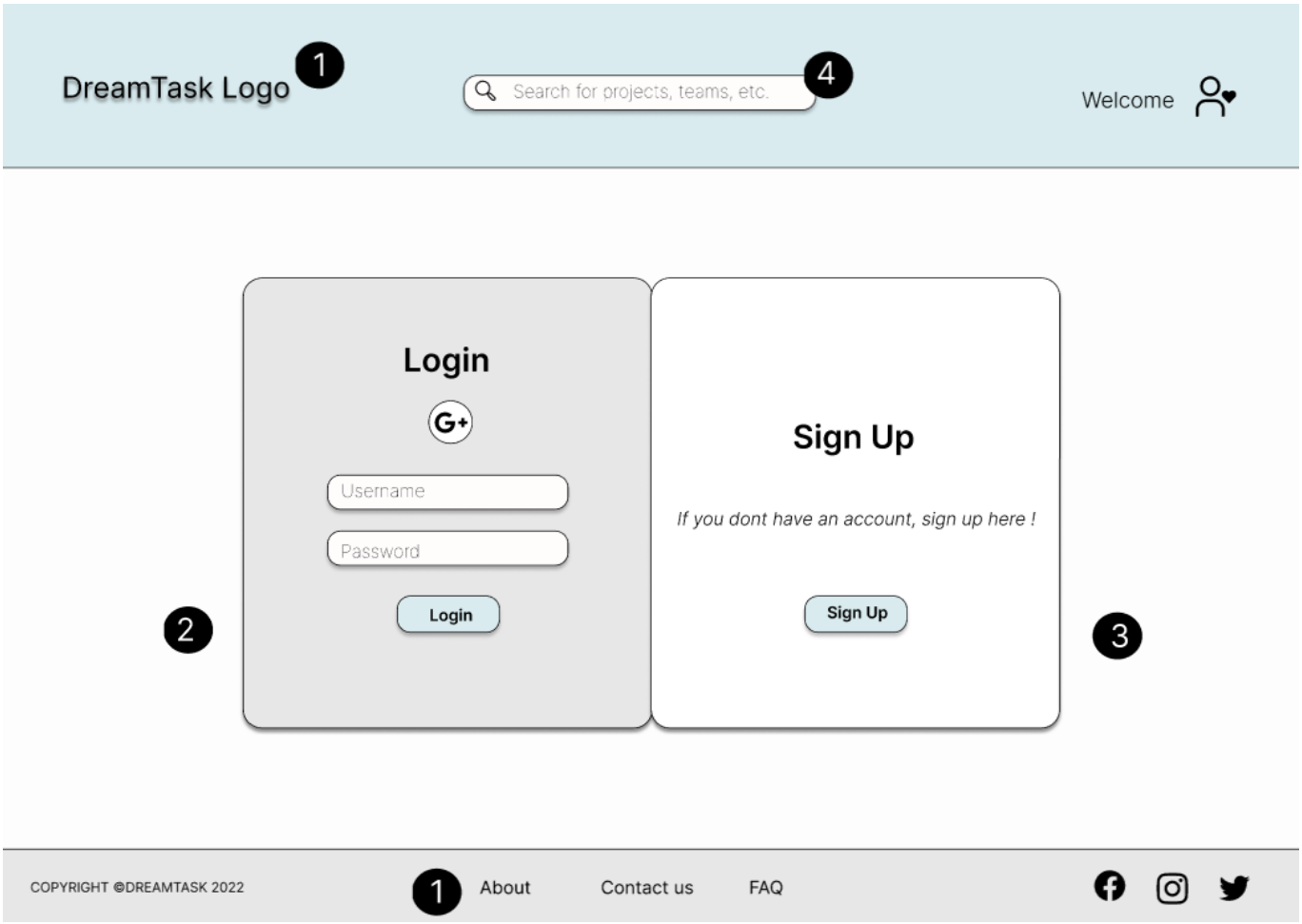


Figure 4: Login Wireframe

1. Navigation - logo as home button and another pages (about, contacts, FAQ)
2. Login area (username and password or using authentication API)
3. Sign Up redirection page area
4. Search Bar for projects, teams, users

2.4 User Tasks Page

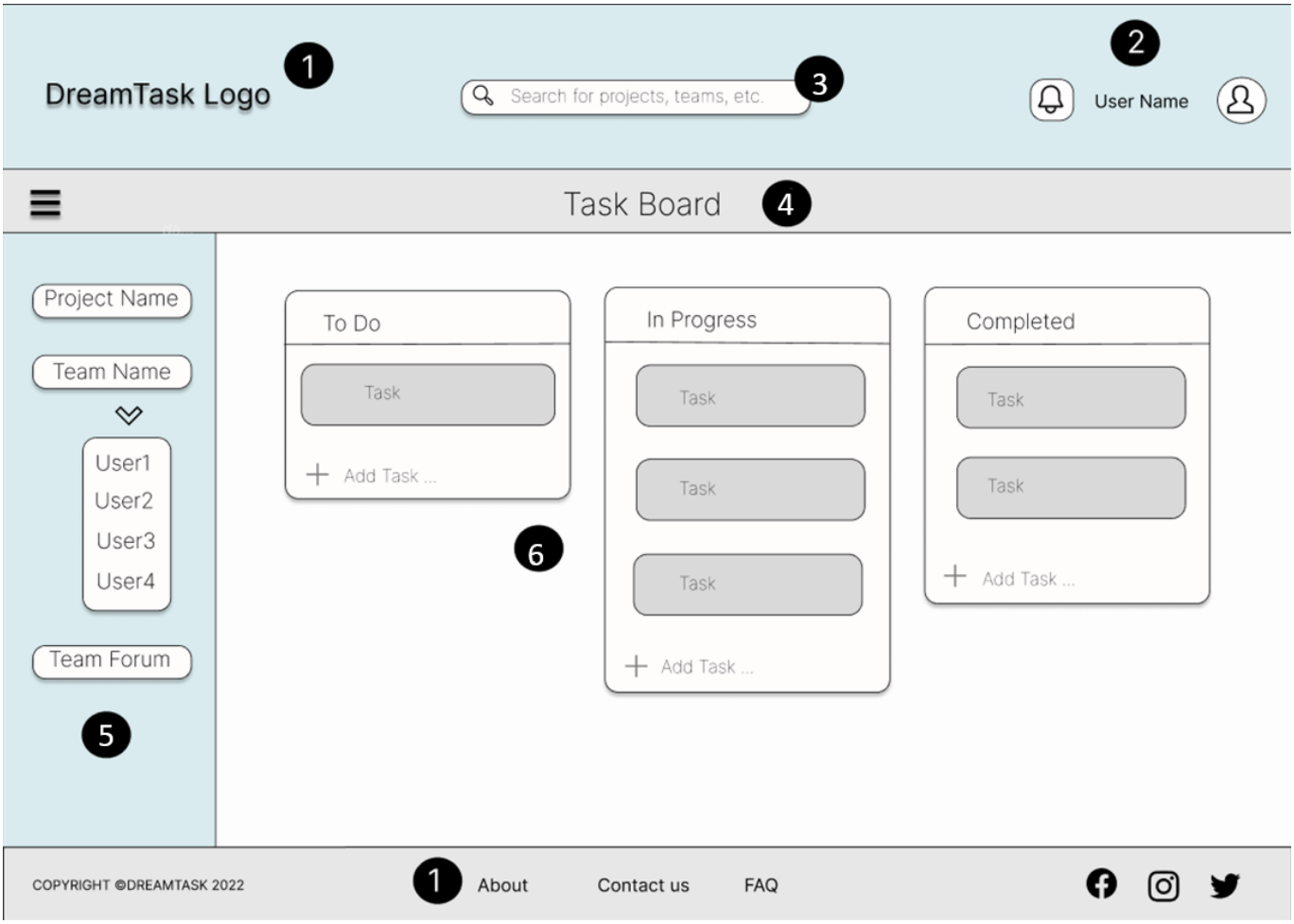


Figure 5: User Tasks Wireframe

1. Navigation - logo as home button and another pages (about, contacts, FAQ)
2. User area (profile and notifications)
3. Search Bar for projects, teams, users
4. Task Board for team tasks management
5. Redirection links to the team, project, team members or message forum (info associated to that team)
6. All the respective Tasks (To Do, In Progress, Completed)

Team

- Alberto Serra up202103627, up202103627@g.uporto.pt
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- Luís Paiva up202006094, up202006094@g.uporto.pt
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